

Chatbot FAQ System Report

Summary

We've implemented a smart, responsive FAQ chatbot to help customers get immediate answers about public swims and lessons.

This system uses cutting-edge natural language understanding to match visitor questions with relevant, pre-approved answers -- even if they don't phrase things exactly as we've written them.

The chatbot is a dynamic upgrade over a traditional FAQ page. Instead of scrolling through long lists, users can simply ask a question in their own words and get an instant reply -- just like talking to a helpful staff member.

How It Works

1. FAQ Database: We maintain a list of approved FAQs in our admin system.
2. User Interaction: Users type questions into the chat widget.
3. Semantic Matching: The chatbot uses AI-powered embeddings to understand the intent behind the question.
4. Confidence Threshold: The system calculates how closely the user question matches an FAQ.
5. Fallback to GPT-4: If no match is found, it generates an answer from structured data using GPT-4.

Benefits Over a Static FAQ Page

- Natural language input vs manual searching.
- Smarter matching even with different phrasing.
- Faster and more focused answers.
- Easier for staff to maintain and update.
- All questions are logged and reviewable.

Monitoring and Improvements

All chatbot queries are stored with timestamps, session keys, and (for FAQ answers) a confidence score. Admins can use this to find unanswered or misunderstood questions and promote them into new FAQs.

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Future Enhancements

- Auto-suggest new FAQs from unmatched chat logs.
- Highlight low-confidence matches.
- Let staff vote on FAQ helpfulness.

Technical Explanation

The system uses OpenAI's embedding model (text-embedding-ada-002) to convert both user questions and stored FAQ entries

into high-dimensional vectors. When a question is submitted, the chatbot embeds it and compares it to all FAQ vectors

using cosine similarity. If the best match exceeds a confidence threshold, the matching answer is returned.

If no good match is found, the chatbot builds a prompt with available structured data (swim times, lessons, term dates, etc.)

and sends it to GPT-4 to generate a natural response. All answers and scores are logged.

The admin has tools to generate embeddings, test matches, and promote queries into FAQs interactively.